

Senior Data Engineer with **11+ years of experience** designing, building, and operating **large-scale, real-time and batch data platforms** across **fintech, consumer marketplaces, and regulated industries**. Proven track record delivering **mission-critical data systems** supporting **high-volume transactions, compliance reporting, and advanced analytics**, including processing **10TB+ daily data** and **1M+ real-time events per day**. Deep expertise in **streaming architectures (Kafka, Spark)**, **lakehouse and warehouse platforms (Databricks, Snowflake)**, **cloud-native data engineering (AWS, Terraform, Kubernetes)**, and **data quality, governance, and security** in PCI-DSS and GDPR-regulated environments. Known for translating complex business problems into **reliable, scalable data solutions**, leading cross-functional collaboration with Risk, Finance, Product, and Operations teams, and driving measurable outcomes such as **50% latency reduction, 40% scalability gains, and 30% query performance improvements**.

WORK EXPERIENCE

DraftKings, Boston, MA

Senior Data Engineer

May 2023 - Present

- Migrated **15+ real-time wagering and betting event pipelines** from **AWS Kinesis to Apache Kafka (MSK)**, powering ingestion of **bets, odds updates, settlements, and payouts**. Built event-driven streaming jobs in **PySpark on Databricks**, handling **late and out-of-order events**, enforcing schema evolution, and guaranteeing **exactly-once processing** for financial accuracy, reducing latency by **50%** and improving scalability by **40%**.
- Designed and operated **Databricks lakehouse pipelines** on **Amazon S3** using **Spark and Delta Lake** to process **10TB+ of daily betting, player activity, and transaction data**. Supported core DraftKings use cases including **risk exposure monitoring, fraud detection, player behavior analysis, and revenue forecasting**, cutting model and analytics deployment time by **40%**.
- Built and optimized **50+ PySpark ETL and streaming jobs** ingesting high-volume wagering and tax data during peak sports traffic. Implemented a **Bronze/Silver/Gold architecture** to separate raw betting events, validated financial records, and analytics-ready datasets used by Risk, Finance, and Trading teams.
- Engineered and maintained **20+ Snowflake schemas** backing **bet settlement, payouts, taxation, and regulatory reporting**. Integrated upstream systems from **Amazon RDS (PostgreSQL)** and **MongoDB**, tuning clustering and query patterns to improve performance by **30%** during high-concurrency reporting windows tied to major sporting events.
- Established **data quality and financial reconciliation frameworks** across streaming and batch pipelines, including freshness checks, volume validation, schema enforcement, and anomaly detection. Ensured **high-trust data** used for settlements, compliance reporting, and executive dashboards where data errors carry regulatory and financial risk.
- Served as **on-call owner** for mission-critical betting data pipelines during **NFL, NBA, and international tournaments**, rapidly diagnosing **Kafka consumer lag, Spark failures, and Snowflake bottlenecks** under extreme traffic spikes to ensure uninterrupted wagering and payout operations.
- Partnered closely with **Risk, Trading, Finance, Compliance, and DevOps teams** to design **secure, auditable data flows** aligned with **PCI-DSS, GDPR, and gaming regulations**. Leveraged **AWS (S3, RDS, Lambda, IAM, CloudWatch)** and **Terraform** to enforce access controls, monitoring, and infrastructure consistency.
- Provided **technical leadership and mentorship** across the data engineering team, leading design reviews for streaming and lakehouse architectures. Drove **cost optimization initiatives** by tuning Spark workloads, cluster sizing, and storage formats, reducing infrastructure spend while maintaining low-latency performance during peak betting periods.

Sweetgreen, Los Angeles, CA

Senior Data Engineer

Apr 2021 - Apr 2023

- Built and optimized **data pipelines in Python and Java** supporting **digital ordering, in-store POS, loyalty, labor, and inventory systems** across **hundreds of Sweetgreen locations**. Orchestrated ETL workflows with **Apache**

Airflow and dbt, and designed **Snowflake** data models to power store-level performance tracking, menu analytics, and leadership reporting while normalizing inconsistent data from restaurant systems and vendors.

- Partnered closely with **Operations, Culinary, Supply Chain, Finance, and Product teams** to translate restaurant workflows into analytics-ready datasets. Maintained **data models, schemas, and data dictionaries** for metrics such as **order throughput, ingredient consumption, labor efficiency, and store profitability**, and built internal **FastAPI services** to expose trusted data for operational tools, tested and documented with **Postman**.
- Established **data governance, quality, and security standards** for operational and financial data flowing from stores, kitchens, and suppliers. Built and maintained Airflow- and dbt-driven workflows on **AWS (S3, RDS, Lambda)** to ensure consistent reporting across restaurants, third-party delivery platforms, and supplier feeds while addressing reconciliation and consistency challenges.
- Developed **real-time streaming pipelines** using **Kafka and AWS Kinesis** to ingest **order events, inventory movements, labor signals, and kitchen activity**. Optimized **PostgreSQL and MongoDB** performance and used **Pandas and PySpark** for transformations, ensuring reliable data flow during **lunch and dinner rush periods** when transaction volume spikes sharply.
- Designed and maintained **data APIs and system integrations** using **FastAPI** to enable seamless data sharing between ordering, inventory, analytics, and reporting platforms. Performed deep operational analysis to support **menu optimization, demand forecasting, labor planning, and supplier performance**, and implemented **Databricks-based predictive analytics workflows** tailored to restaurant operations.
- Provided **technical mentorship** to junior data engineers, promoted best practices in ETL design and data modeling, and collaborated with data scientists to prepare clean, well-structured datasets for forecasting and optimization models. Experimented with **AI-assisted analytics tooling** to accelerate insight generation for restaurant and supply-chain teams.

Robinhood, Menlo Park, CA

Data Engineer

Nov 2017 - Mar 2021

- Built and monitored **high-volume data pipelines** for **orders, trades, executions, and account activity**, processing **1M+ events per day** using **Akka Streams, Python Asyncio, and SQL**, supporting live trading during high market volatility.
- Developed **ETL pipelines with Python and dbt**, integrating data from **AWS S3, Amazon Redshift, Snowflake, and MongoDB**, processing **2TB+ daily** for compliance reporting, risk analysis, and internal analytics.
- Implemented **data security and privacy controls** including **AES-256 encryption, RBAC, and audit logging**, ensuring **PCI-DSS and GDPR** compliance, and contributed to migrating systems from a **monolithic architecture to microservices**.
- Supported **large-scale data migrations and integrations**, including a **5TB production migration** with minimal downtime, validating end-to-end data flows and REST integrations using **Postman**.
- Designed and maintained **data models and schemas** for **BI and regulatory reporting**, adapting structures to evolving financial requirements and supporting dashboards built with **Power BI**.
- Implemented **data validation, reconciliation, and cleansing** using **SQL, Python, and Java**, resolving inconsistencies across **1M+ records** and ensuring alignment between transactional and analytical systems via **REST APIs**.
- Collaborated with **DevOps teams** to automate infrastructure using **Terraform and Jenkins**, containerized data services with **Docker**, and deployed workloads on **Kubernetes** for scalability and reliability.
- Improved **SQL query performance by 30%** through indexing and tuning, supported monitoring and troubleshooting, and contributed to **testing and QA** using **Pytest, Allure, JavaScript automation, Git, Linux, and Bash**.

GrubHub, New York, NY

Python Engineer

Feb 2013 - Oct 2017

- Built and supported **data pipelines and backend services in Python 3** to process **order, restaurant, delivery, and customer data**, helping power Grubhub's food-ordering and delivery operations as the platform scaled across major U.S. cities.

- Developed and maintained **ETL workflows** using **Python, Airflow, dbt, and Talend**, integrating data from **AWS S3, Hadoop, Hive, Spark, and Snowflake** to support reporting and analytics for operations, marketing, and customer experience teams.
- Worked with **large-scale datasets (up to 10TB)** containing order history, delivery events, and restaurant performance metrics, improving data availability and reliability for analytics and internal dashboards.
- Leveraged **Hadoop and Spark** to process and analyze high-volume data, generating insights used to improve **order throughput, delivery efficiency, and inventory/menu availability** across restaurant partners.
- Managed and optimized **data storage solutions** using **Amazon S3, HDFS, and Hive**, supporting secure and efficient access to historical and real-time food delivery data.
- Collaborated with **product, operations, and marketing stakeholders** to align data pipelines with business needs, supporting initiatives around **customer engagement, promotions, and restaurant performance**.
- Built and maintained **data pipelines with Python, Airflow, and dbt**, enabling consistent data ingestion and transformation across multiple internal systems used for food delivery analytics.
- Performed **data analysis and visualization** using **Pandas and Sisense**, delivering reports and insights that helped teams understand ordering patterns and customer behavior.
- Implemented **data access controls and security measures** using **LDAP and Azure**, ensuring safe handling of sensitive customer, restaurant, and transaction data.

SKILLS

- **Data Engineering & Quality:** Data pipelines (batch & real-time), data quality engineering, data validation & QA, data lineage, metadata management, data contracts, dependency management, ETL/ELT optimization, workflow orchestration (Airflow DAGs, backfills, retries, SLAs), data governance & compliance
- **Databases & Storage:** PostgreSQL, MySQL, Snowflake, Amazon Redshift, Azure Synapse, MongoDB, Cassandra, AWS S3, Azure Blob Storage, HDFS, SQL & NoSQL querying
- **Big Data, Streaming & Lakehouse:** Apache Spark, Hadoop, Kafka, AWS Kinesis, Airflow, Apache NiFi, Hive, HDFS, Pig, Talend, Informatica, Delta Lake, Avro, Parquet, ORC, schema evolution
- **Data Modeling & Analytics:** Dimensional modeling, star schema, snowflake schema, ER diagrams, analytical and regulatory schema design
- **Data Visualization & BI:** Tableau, Power BI, Sisense, Looker
- **Machine Learning & Advanced Analytics:** Machine learning pipelines, feature engineering, MLflow, TensorFlow, PyTorch, computer vision (foundational)
- **Cloud Platforms & Infrastructure:** AWS (S3, RDS, Redshift, Lambda, IAM, CloudWatch), Azure (SQL Database, Blob Storage, Data Factory), GCP (BigQuery), Docker, Kubernetes, Terraform, CI/CD pipelines (Jenkins, GitHub Actions)
- **Monitoring, Observability & Security:** Datadog, Splunk, Grafana, ELK stack, data observability, pipeline monitoring, AES-256 encryption, RBAC, GDPR, HIPAA, PCI-DSS
- **Programming & Frameworks:** Python, Java, JavaScript, SQL, FastAPI, Flask, Django
- **Testing & Quality Assurance:** Pytest, test automation, data validation, user acceptance testing (UAT), automated web scraping, data reconciliation
- **Agile & Collaboration:** Agile, Scrum, Kanban, JIRA, Confluence, Git, GitHub, Bitbucket

EDUCATION

Bachelor of Science in Computer Science, Columbia University, New York, NY (Aug 2008 - Oct 2012)